Koen Essers • Master Internship • October 26, 2026

Notes Week 34

a. Bug Report

```
!! @param ACMI An array of the masses in AMU of those species.
ACMI(relevant_species) = chem_isos% W(chem_id(j))
A(relevant_species) = chem_isos% Z_plus_N(chem_id(j))


chem_isos% Z_plus_N = chem_isos% Z + chem_isos% N
chem_isos% W = chem_isos% Z_plus_N + chem_isos% mass_excess*(mev_to_ergs/amu/(clight*clight))
```

b. Summarizing my Results

I will first look at what I think is most relevant in each of my weekly notes.

2 – 3. This week contains a lot of relevant MESA parameters that I use, and *why* I use them. Other than that, there are not really any relevant results.

4. This week shows my first attempts of RLOF mass transfer with MESA. I also check some dredge-up properties of Rees et al. (2024)-like TPAGB models. There are no relevant *new* results.

5. I test different wind prescriptions for single stars. This is relevant for the methods section of my final thesis, but does not contain *new* results. I also compare different opacity tables. Similarly, this is important for the methods, but does not show something *new*.

   I also implement the quasi-adiabatic mass loss limit from Temmink et al. (2023). Again something for the methods. This is also the first week where I try some binary models with RLOF during the TPAGB phase of the donor. Not really any interesting results though, since these models still miss wind-mass transfer and tides.

6. Tried to figure out why models with low R and q fail. Did not really succeed. Contains a good description of ÆSOPUS tables for methods / introduction.

7. Contains detailed information of the overshooting parameters and comparisons with other methods. Also, MesaGrid class version 1. Not very relevant any more.

8.

9.

10.

11.

12.

13.

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

References

Rees, N. R., Izzard, R. G., & Karakas, A. I. (2024) *Thermal pulses with MESA: resolving the third dredge-up*. MNRAS527, 9643.

Temmink, K. D., Pols, O. R., Justham, S., Istrate, A. G., & Toonen, S. (2023) *Coping with loss. Stability of mass transfer from post-main-sequence donor stars*. A&A669, A45.